

LLM 8-dars.

1 ta token o'rtacha 4 bayt bōladi

Ollama → barcha local LMM lar bor

Ketma-ketlik 2

Human Text → Tokenizer → Token ID →
Embeddings → Positional Embedding → Attention
→ FFN → Output

Embedding turlari:

- Positional embedding (wpe)
- Token Embeddings (wte)

wpe → har bir token javlashuvi bōyicha modelga
ingʻa beradi.

wte → tokenlarni ID vektorga aylantiradi

AutoModel → Huggingface dan LLM modellarni
yuklab oladi.

Text
↓

Tokenizer
↓

Token ID
↓

Embedding layer
↓

Transformer